

# **Diffuse Cascade in 3D Navier-Stokes: Time-Resolved Evidence for Triad Equidistribution**

Jonathan Simons. This record (10.5281/zenodo.20183673) is a 2026 exploratory draft. It remains published and citable.<sup>1</sup> This page is the public-facing face of the record. It is not a retraction notice. The original draft follows the errata.

## **Live exploratory stack (cite these, not prize packaging)**

Live exploratory stack:  $\Phi$ -renormalization 10.5281/zenodo.22050974 (not a Clay Navier-Stokes proof); Route C 10.5281/zenodo.22050963 (not a proof of RH); Ring lemma 10.5281/zenodo.22050976 (Goldbach remains conditional on SND); T2 under SND 10.5281/zenodo.22050965 and Bridge\* stay named derived objects; inverse-GCD note 10.5281/zenodo.22050962 is a restricted Rayleigh bound, not a full-spectrum floor.

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<sup>1</sup> August 2026 prize-claim language walked back; see errata below / page 2 of this file and status note 10.5281/zenodo.22050978.

## **Errata: prize-claim language walked back**

The pages after this one are the original 2026 draft, unchanged. They are on the record so the walk-back is auditable. The following prize-claim language in that draft does not stand.

### **Why the August 2026 action happened**

In August 2026, prize-claim language on several 2026 records was walked back. The walk-back was necessary: those drafts packaged Navier-Stokes, the Riemann hypothesis, and Goldbach as closed, and that packaging does not hold. The walk-back was executed badly. Public titles were stamped on 21 August 2026 so the landing page looked like a crime scene. The files were never unpublished, never tombstoned, and never taken off open access. This pack restores the public face: a clean title, an honest first page, and the errata on page 2.

### **Prize-claim language that does not stand**

Clay Statement B / Navier-Stokes millennium packaging. The June drafts treated a regularity claim as closed. That language is walked back. The  $\Phi$ -renormalization note is the live fluids object.

Quantum Lens / Q6 Goldbach as a theorem. Goldbach stays conditional on SND. Q6 is not a proof.

Montgomery-Dyson / pair correlation as a finished RH argument. That packaging is walked back, including the May 14 "Coincidence Resolved" draft. Live RH work is exploratory Route C and the Track B Möbius-GCD attack (obstruction, not a proof).

Three-in-one /  $\text{SND} \equiv \text{GNC} \equiv \text{Bridge}$  as closures. Those identities were used as if they closed NS, RH, and Goldbach. They do not. The June drafts that argued that way remain published; the prize language is walked back.

Full-spectrum  $\lambda_{\min}(Q_N) > -1/2$  as a RH proof. That bound is not a proof of RH.  
Inverse-GCD  $1/\text{gcd}$  is not the live Route C operator.

This page is the errata. It is not a tombstone. No Millennium problem is claimed. Domain Architect is a local FRA classifier (inquiry). ChatVault is search. Neither wrote these Zenodo titles.

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# Diffuse Cascade in 3D Navier-Stokes: Time-Resolved Evidence for Triad Equidistribution

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## Abstract

We present time-resolved numerical evidence that triad amplitudes in the three-dimensional incompressible Navier-Stokes equations remain equidistributed across spectral shells and time. Across resolutions  $N=32, 64, 128$ , no concentration events are observed under stringent diagnostics. The minimum amplitude equidistribution (AET) remains above 0.308, the scaling exponent remains near zero, and no helical sector exhibits persistent dominance. These results indicate a diffuse cascade in which energy transfer is continuously distributed across triads, with no transient or sustained concentration in the tested regimes.

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## 1. Introduction

The problem of global regularity for the three-dimensional incompressible Navier-Stokes equations hinges on whether nonlinear interactions can concentrate energy into small subsets of modes or triads, potentially leading to finite-time blowup. This work investigates that possibility directly by analyzing the structure and distribution of triad interactions in spectral space. Rather than bounding norms indirectly, we examine whether energy transfer concentrates or remains distributed across interacting triads over time.

## 2. Methodology

We use a pseudo-spectral Navier-Stokes solver with triad-level diagnostics. For each snapshot and spectral shell  $j$ , we compute:

- Amplitude Equidistribution:  $AET_j(t) = N_{eff,j}(t) / N_j$
- Scaling exponent:  $AET_j(t) \sim C(t) \cdot 2^{-\alpha(t) \cdot j}$
- Concentration diagnostics: top 1% triad mass fraction, helical sector dominance, time-persistent collapse detection

A "danger event" is defined as any snapshot where  $\text{AET}_j(t) < 0.05$ , top 1% mass  $> 30\%$ , or  $\alpha(t) > 1$ .

### 3. Results

#### Resolution Summary

| Resolution | Danger Events | max $\alpha(t)$ | min AET(t) |
|------------|---------------|-----------------|------------|
| N = 32     | 0             | 0.187           | 0.316      |
| N = 64     | 0             | 0.061           | 0.319      |
| N = 128    | 0             | 0.066           | 0.308      |

#### Observed Properties

- No concentration events observed across any resolution
- Top 1% triad mass  $\leq 10.6\%$
- No persistent helical sector dominance
- AET remains stable across shells and time
- Scaling exponent remains near zero

### 4. Time Evolution

At all resolutions, the system exhibits stable AET across all snapshots, no transient collapse events, no sustained concentration behavior, and no shell-level drift toward localization. The scaling exponent  $\alpha(t)$  remains near zero at all times, indicating that equidistribution persists across spectral shells dynamically — not just on average.

### 5. Interpretation

These results demonstrate that triad amplitudes remain broadly distributed, no subset of triads carries dominant energy transfer, no coherent structure emerges to sustain concentration, and energy is continuously redistributed across interactions. In particular: energy transfer remains distributed across triads at all times. No transient or sustained concentration mechanism is detected.

### 6. Discussion

The observed behavior is consistent with a diffuse cascade, in which energy transfer occurs through a distributed network of interactions rather than concentrated pathways. This challenges the viability of blowup mechanisms that rely on triad-level concentration, persistent alignment, or localized amplification. Instead, the cascade appears structurally resistant to concentration.

## 7. Limitations

These results are based on finite-resolution simulations. While consistent across multiple resolutions  $N=32, 64, 128$ , the asymptotic behavior as  $j \rightarrow \infty$  remains an open question requiring higher-resolution studies.

## 8. Conclusion

We provide numerical evidence that triad amplitudes remain equidistributed across shells and time, no concentration events occur across tested resolutions, and energy transfer is dynamically distributed rather than localized. This supports a picture of the Navier-Stokes cascade as a diffuse, non-concentrating system within the tested regimes.